

LAB7

IT24100956

```
[kali@kali] ~/Desktop/SE LAB/Exercise01
mpicc -Wall -O2 -o sum_bcast sum_bcast.c

[kali@kali] ~/Desktop/SE LAB/Exercise01
$ mpirun -p 4 ./sum_bcast

[kali][58986,1,0][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,0][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,0][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,1][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,1][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,3][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,3][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,3][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,2][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,2][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][58986,1,2][...]/.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
Root filled array with values 1 to 1000000
Rank 0: summed indices [0, 250000] => local_sum = 312500125000
Rank 1: summed indices [250000, 500000] => local_sum = 937500125000
Rank 3: summed indices [750000, 1000000] => local_sum = 218750125000
Rank 2: summed indices [500000, 750000] => local_sum = 156250125000

[Bcast] Total sum = 5000000500000
[Bcast] Expected = 5000000500000
[Bcast] Correct? YES
[Broadcast] time = 0.0051 sec

[kali@kali] ~/Desktop/SE LAB/Exercise01
```

```

kali@kali: ~/Desktop/SE LAB/Exercise2
└─ cd Exercise2

kali@kali: ~/Desktop/SE LAB/Exercise2
└─ mpirun -wall -Q --sum_scatter sum_scatter.c

m1w@ ~4 /sum_scatter
[kali][5784,1,2] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,3] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,2] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,2] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,3] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,3] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,0] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,1] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,0] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,1] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,1] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][5784,1,0] .../.../opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
Rank 0: filled array with values 1 to 1000000
Rank 0: summed indices [0, 250000] => local_sum = 31250125000
Rank 2: summed indices [500000, 750000] => local_sum = 156250125000
Rank 3: summed indices [750000, 1000000] => local_sum = 218750125000
Rank 1: summed indices [250000, 500000] => local_sum = 93750125000

[Scatter] Total sum = 500000500000
[Scatter] Expected = 500000500000
[Scatter] Correct? = YES
[Scatter] Time = 0.0011 sec

kali@kali: ~/Desktop/SE LAB/Exercise2
└─ s5

```

Exercise 03

```
(kali@kali) ~/Desktop/SE LAB
$ cd Exercise03
(kali@kali) ~/Desktop/SE LAB/Exercise03
$ mpicc -Wall -D2 -o sum_gather sum_gather.c
mpirun -np 4 ./sum_gather
[kali][31562,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,3] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph [kali][31562,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable
to find reachable pairing between local and remote interfaces
[kali][31562,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,0] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,0] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][31562,1,0] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
Root filled array with values 1 to 1000000
Rank 0: summed indices [0, 250000) => local_sum = 31250125000
[gather] Total sum = 500000500000
[gather] Expected = 500000500000
[gather] Correct? = YES
[gather] Time = 0.0011 sec
Rank 2: summed indices [500000, 750000) => local_sum = 356250125000
Rank 1: summed indices [250000, 500000) => local_sum = 93750125000
Rank 3: summed indices [750000, 1000000) => local_sum = 218750125000
(kali@kali) ~/Desktop/SE LAB/Exercise03
```

Exercise 04

```
(kali@kali) ~/Desktop/SE LAB
$ cd Exercise04
(kali@kali) ~/Desktop/SE LAB/Exercise04
$ mpicc -Wall -D2 -o sum_reduce sum_reduce.c
mpirun -np 4 ./sum_reduce
[kali][319138,1,0] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,2] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,0] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,3] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
[kali][319138,1,1] ./opal/mca/btl/tcp/btl_tcp_proc.c:266:mca_btl_tcp_proc_create_interface_graph Unable to find reachable pairing between local and remote interfaces
Root filled array with values 1 to 1000000
Rank 2: summed indices [500000, 750000) => local_sum = 356250125000
Rank 0: summed indices [0, 250000) => local_sum = 31250125000
Rank 3: summed indices [750000, 1000000) => local_sum = 218750125000
Rank 1: summed indices [250000, 500000) => local_sum = 93750125000
[Reduce] Total sum = 500000500000
[Reduce] Expected = 500000500000
[Reduce] Correct? = YES
[Reduce] Time = 0.0013 sec
(kali@kali) ~/Desktop/SE LAB/Exercise04
```


Exercise 07

1. COMPARISON TABLE:					
Program Name	Collectives Used	Memory Allocation Per Process	Root Needs Manual Sum Loop?	Where Final Result is Available	Communication Complexity (Data Dist. / Collection)
1. sum_bcast	MPI_Bcast	Every process allocates FULL N	Yes (manual MPI_Recv loop)	Root only	$O(N \log P)$ / $O(P)$ linear point-to-point
2. sum_scatter	MPI_Scatter	Only root allocates N; others allocate N/P	Yes (manual MPI_Recv loop)	Root only	$O(N)$ linear scatter / $O(P)$ linear point-to-point
3. sum_gather	MPI_Scatter, MPI_Gather	Only root allocates N + P; others allocate N/P	Yes (loops over all sums)	Root only	$O(N)$ / $O(P)$ tree-based gather
4. sum_reduce	MPI_Scatter, MPI_Reduce	Only root allocates N; others allocate N/P	No (computed by MPI_Reduce)	Root only	$O(N)$ / $O(\log P)$ tree-based reduction
5. sum_allreduce	MPI_Scatter, MPI_Allreduce	Only root allocates N; others allocate N/P	No (computed by Allreduce)	All processes	$O(N)$ / $O(\log P)$ recursive-doubling / butterfly
6. sum_scan	MPI_Scatter, MPI_Scan	Only root allocates N; others allocate N/P	No (computed by MPI_Scan)	Different per rank (prefix running sum up to each rank; total on last rank)	$O(N)$ / $O(\log P)$ parallel prefix tree

2. EXECUTION TIME BENCHMARKS (2, 4, AND 8 PROCESSES):			
Benchmark tests were performed across all 6 implementations using 1,000,000 integers. Execution times (in seconds) averaged across multiple runs:			
Implementation	np = 2	np = 4	np = 8
1. sum_bcast	0.0002 s	0.0051 s	0.0078 s
2. sum_scatter	0.0019 s	0.0008 s	0.0034 s
3. sum_gather	0.0031 s	0.0022 s	0.0025 s
4. sum_reduce	0.0026 s	0.0018 s	0.0019 s
5. sum_allreduce	0.0027 s	0.0019 s	0.0021 s
6. sum_scan	0.0028 s	0.0019 s	0.0022 s

Exercise 08

```
1 CC = mpicc
2 CFLAGS = -Wall -O2
3 NP = 4
4
5 TARGETS = sum_bcast sum_scatter sum_gather sum_reduce sum_allreduce sum_scan
6
7 .PHONY: all run clean
8
9 all: $(TARGETS)
10
11 sum_bcast: sum_bcast.c
12 $(CC) $(CFLAGS) -o $@ $<
13
14 sum_scatter: sum_scatter.c
15 $(CC) $(CFLAGS) -o $@ $<
16
17 sum_gather: sum_gather.c
18 $(CC) $(CFLAGS) -o $@ $<
19
20 sum_reduce: sum_reduce.c
21 $(CC) $(CFLAGS) -o $@ $<
22
23 sum_allreduce: sum_allreduce.c
24 $(CC) $(CFLAGS) -o $@ $<
25
26 sum_scan: sum_scan.c
27 $(CC) $(CFLAGS) -o $@ $<
28
```